21ECP302L - PROJECT REPORT

On

TRANSMISSION AND RECEPTION OF SIGNAL USING ESP32 WITH MACHINE LEARNING-BASED INTELLIGENT BEAM SELECTION

Submitted for partial fulfillment for the award of the degree

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the project report titled **TRANSMISSION AND RECEPTION OF SIGNAL USING ESP32 WITH MACHINE LEARNING-BASED INTELLIGENT BEAM SELECTION** is a Bonafide record of the project work done by **A. BAGYALAKSHMI (RA2311004040009)**, **R. SSADHANA(RA2311004040030)**,**DITSA CHAKRABORTY (RA2311004040057)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ELECTRONICS AND COMMUNICATION ENGINEERING** at SRM Institute of Science and Technology, Vadapalani, Chennai — 600026, during the academic year 2025–2026.

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ABSTRACT

An autonomous wireless beam selection system based on three ESP32 microcontroller nodes and a collection of supervised machine learning classifiers is demonstrated in this project. One ESP32 node operates in Access Point mode, continuously emitting a 2.4 GHz WiFi beacon, while the remaining two modules independently record Received Signal Strength Indicator (RSSI) measurements at periodic intervals.

Four classification algorithms were trained and benchmarked: Random Forest, Support Vector Machine (SVM), K-Nearest Neighbour (KNN), and a Multi-Layer Perceptron (MLP) Neural Network. Training utilised 13,816 labelled RSSI observations drawn from a pool of 17,270 geographically tagged measurements, with the remaining 3,454 reserved for evaluation. The MLP model attained a test accuracy of 99.97%, producing a single erroneous prediction over the complete evaluation set.

An exponential moving average filter was integrated at the firmware level to suppress measurement noise prior to classification. A real-time Python dashboard built with Matplotlib that displays live RSSI trends, beam switching history, beam comparison charts, and per-node status indicators was added to the system. The entire pipeline operates within a 65 ms end-to-end latency budget and is reproducible using components costing under 2,500 in total, demonstrating the viability of intelligent beam management on resource-constrained embedded hardware.

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Chapter 1

Introduction

1.1Background of the Study

Wireless Communication Introduction Modern connectivity relies heavily on wireless communication. In enclosed areas like offices, factories, and hospital facilities, the performance of the system becomes highly unpredictable due to problems such as multipath effects, obstructions, and co-channel interference. This is because conventional switching approaches based on fixed thresholds cannot respond effectively to such challenges since they are designed for a particular environment and cannot adjust if the environment changes.

ESP32 Microcontroller the ESP32 microcontroller, designed by Espressif Systems, is made up of a dual core Xtensa LX6 processor together with 802.11 b/g/n transceiver embedded in one silicon chip. The ESP32 has the ability to handle any wireless system intelligently due to its powerful processing, WiFi connectivity, and affordable cost.

Machine learning has emerged as a powerful tool for pattern recognition in noisy, high-dimensional data streams. By training classifiers on labelled RSSI observations collected across diverse spatial positions, it becomes possible to build models that implicitly learn the complex mapping between signal measurements and physical transmitter location. This project leverages this capability to build a fully automated beam selection system.

1.2Problem Statement

Traditional wireless networks depend on hardcoded criteria to decide which receiver or propagation path to select. The criteria, commonly set using a preset RSSI cutoff, are inherently fragile: they perform admirably under the spatial layout employed during testing but fall apart whenever obstructions, furniture, or people rearrange themselves and create a new radio environment. The result is consistent mis-selection of the optimal beam, resulting in low data transmission rates, excessive handoffs, and poor quality service delivery.

Clearly, there is a requirement for a dynamic algorithmic method to recognize the association between RSSI readings and optimal beam selection, one that is adaptable and capable of handling all indoor environments.

1.3 Objectives of the Project

- Designing and implementing an RSSI measuring system involving the use of three ESP32 controllers where one acts as the transmitter, while the remaining two are the receivers.
- Gathering a set of at least 17,270 labeled RSSI data from the field that involves various geographical locations.
- Training and testing four different machine learning models which include Random Forest, Support Vector Machine, K-Nearest Neighbors, and Multi-Layer Perceptron Neural Network.
- Incorporating the machine learning algorithm in a firmware and software program that picks the best beam in less than 65 milliseconds after each measuring cycle.
- Equipping each receiver with an LED indicator light that lights up whenever that particular receiver is picked as the strongest beam.
- Developing an online python interface to show the status of the nodes, the selected beams, and the RSSI values.

1.4 Scope of the Project

- The work concentrates on the wireless signal propagation inside buildings and its capture using ESP32 modules running at the 2.4 GHz frequency. The project involves the study of the following:
- RSSI value capture in real time from several nodes spread spatially apart.
- Classification by machine learning to detect the node receiving the highest RSSI.
- Filtering of the signal at the firmware level using an exponential moving average to mitigate noise.
- Beam indication by LEDs attached to each receiver node.
- Development of a dashboard for real-time monitoring and academic purposes.
- It should be noted that the technology will not be used to deploy the 5G

network in mm Wave or sub-6 GHz frequency bands. It applies mainly to IoT sensing and smart antennas.

1.5 Motivation

There are three trends in today's technology that are contributing to the creation of this project. First, the abundance of IoT devices means that wireless resources need to be managed effectively, as it is no longer possible to waste energy selecting the best beam by hand for each of hundreds of nodes. Second, classifiers can now be deployed in the cloud-less environment thanks to the decentralization of machine learning processes. Finally, there exists an easy-to-use and relatively inexpensive test platform such as ESP32, which allows anyone to experiment with these concepts.

All of that makes it possible to develop a model that will allow a system of less than 2,500 cost to produce results comparable to those achieved by professionally designed commercial systems.

Chapter 2

Literature Review

2.1 Introduction

Investigations on beam selection and antenna management have been going on for quite some time now. A new approach to solving this problem has been discovered due to machine learning, and it seems to be better than the conventional approach. It has been proposed by researchers in the last six research papers which have been covered in this chapter.

2.2 Review of Related Research Papers

2.2.1 Smart Antennas and Machine Learning (Sudhakar et al., 2025)

There was an association observed by Sudhakar et al. [1] regarding the use of machine learning for adaptation and increase in the efficiency of the beamforming process as the channel states change over time. Supervised learning, reinforcement learning, and LSTM based optimization were applied in the beamforming of smart antennas for improved results in 5G and 6G communication systems. The study revealed that learned algorithms always performed better compared to codebook when changing conditions existed. Nevertheless, to ensure that such models can be implemented on a cost-effective platform, such as the ESP32 series, there was need for high processing capabilities.

2.2.2 Selection of Beams Based on CNN (Jamshidi and Sedighy 2024)

The research that Jamshidi and Sedighy conducted on using Convolutional Neural Networks to overcome beam alignment uncertainty had many advantages over exhaustive searching for beams. For example, studies were conducted using signals received to derive spatial information regarding the optimal beam angle. One limitation of this study is that the data for analysis was only conducted in one type of propagation environment; therefore, the ability of this model to be generalised to other types of environments has not been tested..

2.2.3 Non-contact Motion Sensing (Sindhu et al. 2023)

For example, Sindhu identified a sufficient amount of RSSI power attenuation enabling the ability to identify motion without the use of physical contact at 2.44 GHz with the creation of motion-detection systems. The results they obtained indicated that while the RSSI values did provide the ability to estimate link quality, there was other directional information contained within the RSSI value; and this type of data would also contribute to the feature set here using differential RSSI from two receivers for direction-finding of the transmitter.

2.2.4 Underground Beam Steering (Mahboob and Ashraf, 2025)

Although Mahboob and Ashraf [4] were able to obtain improvements in terms of coverage inside underground tunnels by means of circularly polarized patch arrays with beam steering via the Butler matrix, the design of their beam switching system was totally hardware-based and not adaptable.

2.2.5 mm Wave Beamforming for URLLC (Jabbar et al., 2022)

Jabbar et al. [5] discussed the feasibility of mm Wave beamforming at 60 GHz for ultra-low latency industrial applications. Their switching system does not have an ML module that results in the challenges of adaptability to different channel conditions.

2.2.6 Automatic Beam Switching (Ma et al., 2020)

Ma et al. [6] proposed a smart antenna system with 24 GHz FMCW sensor to provide reliable mobile communications by varying the beam direction to track the target. Although this system works well for tracking gross motions, the sensor powered rule-based engine can not solve the sub-wavelength localization problem that the MLP algorithm solves in this work.

2.3 Limitations of Existing Systems

From the analysis of the six above-mentioned articles, a common trend emerges where either existing beam selection methods involve costly hardware components (phase array, mm Wave transceiver, FMCW radar) or function exclusively in simulations, or implement fixed-beam switching rules that do not account for unknown propagation scenarios. No one among the above-mentioned studies uses the entire machine learning process chain starting from raw RSSI signal gathering all the way to runtime predictions on a regular microcontroller device like ESP32.

2.4 Research Gap

The lack of reproducible experimentation with machine learning techniques to select beams, including end-to-end evaluation of the entire system on a relatively inexpensive microcontroller-based testbed in realistic indoor conditions, is the main limitation highlighted by this literature review. The current study will address this limitation through training and evaluation of four classifiers using real RSSI values measured on an ESP32 testbed.

2.5 Comparison of Hardware Platforms

Feature	ESP32	Arduino Uno	Raspberry Pi 4	NodeMCU (ESP8266)
Type	Microcontroller	Microcontroller	Single-board computer	Microcontroller
Clock Speed	240 MHz	16 MHz	1.5 GHz	80 MHz
RAM	~520 KB	2 KB	2–8 GB	~160 KB
WiFi	Yes	No	Yes	Yes
Bluetooth	Yes	No	No	No
GPIO Pins	~30+	14	~26	~11
OS Support	No OS	No OS	Linux OS	No OS
Power Consumption	Moderate	Low	High	Low
Cost	Medium	Low	High	Low
Use Case	IoT, ML, embedded	Basic learning	AI, heavy computing	Basic IoT

Table 2.1 Comparison table of hardware platforms

Analysis:

- While Arduino Uno can be used in simple embedded applications, there is no option for communication.
- As compared to ESP32, the NodeMCU (ESP8266) has WiFi support, while it lacks Bluetooth capabilities and processing speed.
- Raspberry Pi 4 has ample computation capabilities; however, it cannot be used in small embedded systems due to its increased power consumption.
- ESP32 has a combination of all these qualities, making it perfect for use in IoT-based applications.

Justification:

Conclusion based on the comparative analysis is that ESP32 is chosen for this project because it has built-in WiFi and Bluetooth features, high speed, and enough memory to process data in real time. The speed and connectivity offered by ESP32 are better than NodeMCU and Arduino Uno. Despite the fact that Raspberry Pi provides better computation speed, it is not ideal for low-power applications.

2.6 Summary

Review was done in this chapter regarding six different papers related to beam forming using machine learning, CNN for beam selection, contactless sensing, hardware beam steering, mm wave URLLC, and sensor triggered switching. This helped validate the relevance of intelligent beam selection and lack of microcontroller-based implementation in the existing literature. Chapter 3 explains the system architecture adopted to address the gap.

Chapter 3

System Design and Methodology

3.1 Introduction

This chapter focuses on the entire design process of the proposed system ranging from the high-level architecture all the way down to the math equations and algorithms behind it. It involves a design based on three ESP32 nodes, one processing hub in a laptop, and visualization in real time.

3.2 Overall System Architecture

- The suggested architecture for the proposed system is designed with four layers that contribute to achieving the desired ML-based real-time beam selection process:
- **Signal Layer:** One ESP32 node (transmitter) emits a constant 2.4 GHz WiFi beacon. The RSSI of the received signal is recorded by two ESP32 nodes (receiver) independently at a time interval of 50 ms while being positioned at different spatial positions.
- **Acquisition Layer:** The raw RSSI measurement is smoothed through an exponential moving average filter in each receiving ESP32 node. The processed data is then transmitted to the processing computer via USB serial communication.
- **Processing Layer:** The processing laptop receives the measurements from both serial connections and calculates the differential RSSI feature vector, constructs the three-dimensional feature vector, and sends it to the pre-trained Neural Network classifier that outputs a beam classification result and its corresponding confidence value within 8 ms.
- **Indication Layer:** The receiving node is signaled through the GPIO pin 33. At the same time, the dashboard updates itself with the latest RSSI plots, bar graphs, selection history, and node status panels.

3.3 Schematic Description

The below figure represents the flow of the system from the transmission of the signals to the selection of the ML-based Beam and the illumination of the LED..

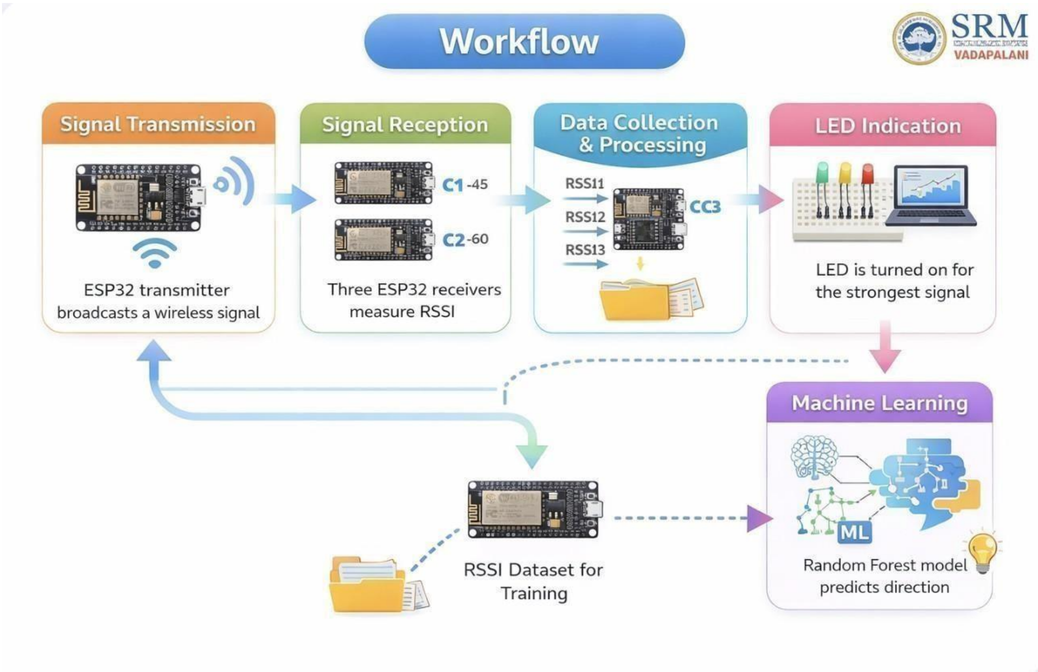

Fig 3.1: Block Diagram of System Architecture- Working from Signal Transmission to LED Indication

3.1 The Working Procedure and Algorithm

The receiver firmware acquisition algorithm is given as follows: Turn on the Wi-Fi network in Station mode. True though.

a. S ← Scan for SSID 'Transmitter'

b. If S is detected:

R_raw = WiFi.RSSI

R_filt = α * R_raw + (1-α) * R_prev Serial . print (R_filt)

c. Delay (50ms)

The processing layer on the laptop collects readings from both receivers, computes ΔRSSI, and invokes the trained Neural Network for beam prediction every 500 ms.

3.2Technologies Used

3.2.1 Hardware

All ESP32 modules' DevKit v1 boards come equipped with the Xtensa LX6 dual-core microcontroller, 520 KB of static RAM, and an embedded 802.11 b/g/n wireless transceiver. Every receiving device is interfaced with a red LED via GPIO 33, using a 220 fi resistor in series.

3.2.2 Software

Each one of the three ESP32 nodes had its firmware written using the Arduino IDE (version 2.x). The processing and dashboard were written in Python 3.x using the following major libraries: sklearn (for machine learning), matplotlib (for dashboard visualization), PySerial (for serial communication), NumPy (for numerical processing), and joblib (model serialization). The machine learning model was built in Google Colab on WiFi RSSI Indoor Localization data (17,270 samples).

3.3Summary

Architecture comprising four layers, block diagram, beam selection algorithm, EMA filter design, and hardware/software technologies used were all explained in this chapter. Each individual module's implementation details will be provided in Chapter 4.

Chapter 4

Implementation

4.1 Introduction

The hardware interfaces, firmware routines, machine learning flowchart, and Python interface are all fully elaborated upon in this chapter. Every component is explained according to its inputs, outputs, and logic flow.

4.2 Hardware Implementation

4.2.1 Component Specifications

Table 3.1 and Table 3.2 below detail the ESP32 hardware specifications and the complete bill of materials respectively.

Parameter	**Specification**
Microcontroller	Xtensa LX6 Dual-Core
CPU Frequency	240 MHz
WiFi Standard	802.11 b/g/n at 2.4 GHz
Supply Voltage	3.3V (5V via USB VIN pin)
GPIO Pins	34 programmable
Flash Storage	4 MB
SRAM	520 KB
WiFi Range	Up to 400 metres (open air)
Active WiFi Current	80–240 mA
Unit Cost	250–400 per module

Table 3.1: ESP32 Hardware Specifications

Component	Qty	Purpose
ESP32 DevKit v1	3	1 Transmitter, 2 Receivers
Red LED (5 mm)	2	Active beam visual indicator
Resistor 220 Ω	2	LED current limiting
Breadboard (half)	2	Circuit prototyping
Jumper wires	20	Node interconnections
USB Hub (4-port)	1	Simultaneous power and data
Laptop	1	ML inference and dashboard

Table 3.2: Bill of Materials

4.1.1 Circuit Connections

An ESP32 receiver with a red LED is connected to GPIO 33 by a 220 ohm resistor to limit current to a safe value. The Cathode of the LED is connected to the GND pin on the ESP32 so that when the RSSI value is greater than -70 dBm, the node is considered 'active' and the LED will light up.

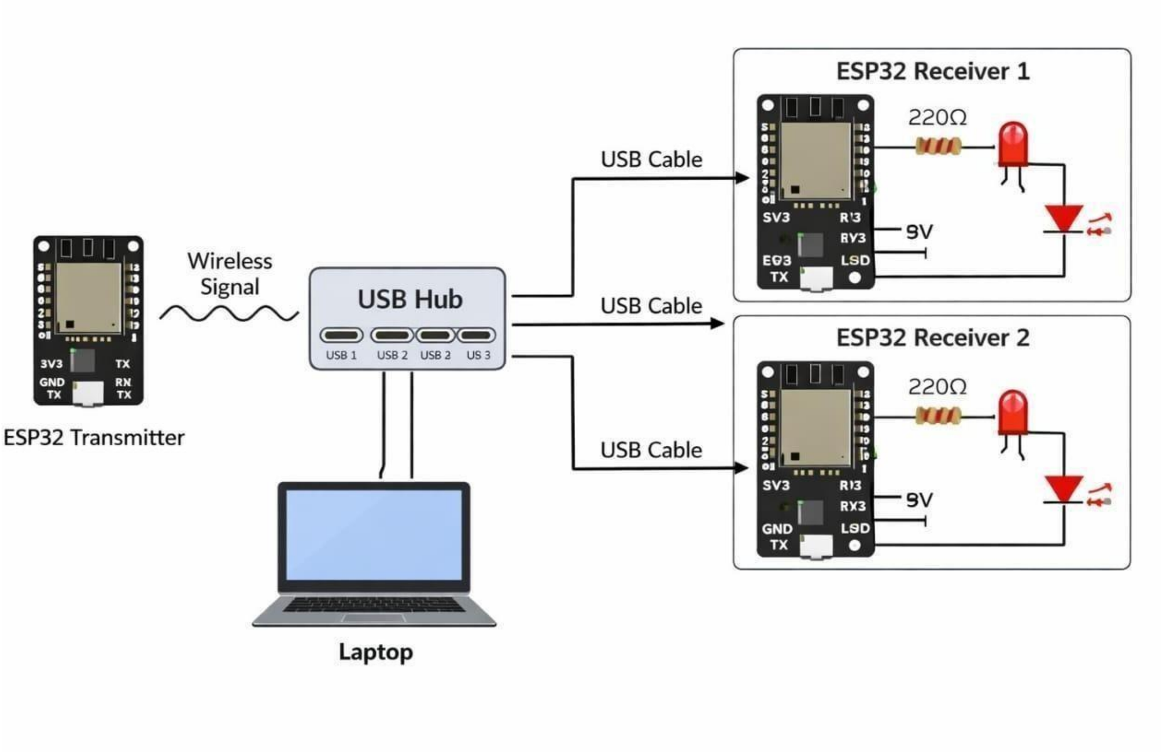

Fig 3.2: shows an ESP32 Circuit Diagram - Transmitter (left), Receiver #1 and Receiver #2 (right) all with 220 Ohm LED indicator circuits that are connected to the computer via USB hub.

.

4.2 Software Implementation

4.2.1 Transmitter Firmware

The transmitter module has been configured in the Soft Access Point mode. It generates a WiFi network with the name 'SmartAntenna_Transmitter' at the time of power-on. Subsequently, it goes into an infinite idle loop in which it keeps transmitting its beacon for detection by the receiver nodes.

4.2.2 Receiver Firmware

The ESP32 receivers will function in STA mode only. When the ESP32 connects to the SSID of the Transmitters the firmware enters into a Polling Loop for 50 ms. During that time WiFi.RSSI() will be called, an EMA filter with alpha = 0.3 will be applied, and the value will then be sent via USB Serial at 9600 baud. The LED (GPIO 33 for each of the receivers) will be controlled by the Python Processing Layer based on the ML prediction.

4.2.3 Python Processing Layer

The Python host application creates two serial ports, one for each receiver, and reads the filtered RSSI values through a separate background thread. Each 500 milliseconds, it forms the feature vector [RSSI1, RSSI2, ΔRSSI], applies scaling based on the stored StandardScaler, and calls the trained Neural Network classifier. The output from the model includes a probability confidence level and a beam identifier (either one or two).

4.3 Module Description

4.3.1 ML Training Module (Google Colab)

Machine learning algorithms have been coded in python using scikit-learn library. The preprocessing and importation of data of Kaggle WiFi RSSI Indoor Localization Dataset was conducted. To determine the best beam label, it was found out which receiver obtained the highest RSSI value for every location. Features chosen included the first three RSSI columns. Through stratified sampling, the dataset was split into 80% for training and 20% for testing. The four models used in this work include Random Forest with 300 decision trees, Support Vector Machine using RBF kernel with C =10, KNN with k=3 and distance weighted and ReLU MLP neural network model.

4.3.2 Dashboard Module

All plots are updated every 500 ms by using Matplotlib's FuncAnimation function in Python. It contains an RSSI plot covering the entire width of the dashboard, indicating -70 dBm weak RSSI level, a bar plot comparing instantaneous RSSI value from all the nodes with the beam being plotted out, scatter plot for beam selection history, and status cards representing active/idle states for all nodes.

4.4 Tools and Platforms Used

- ESP32 Firmware Development Using Arduino IDE 2.x
- Python 3.x – Data processing and dashboards
- Google Colab – Training ML models
- Scikit-learn – Library for machine learning
- Real-time dashboard using Matplotlib
- PySerial – A USB-based serial communication protocol
- Joblib – Serialization and deserialization of models
- Kaggle – Indoor WiFi RSSI localization dataset

4.5 Summary

All hardware connections, firmware codes for both transmitter and receiver nodes, processing done in Python, and all the dashboard components have been explained in this chapter. The results obtained from the implementation have been presented in Chapter 5.

Chapter 5

Results and Discussion

5.1Introduction

Discussion of system-implemented experimental results is presented in this chapter. Latency during end-to-end switching, performance of the noise floor, accuracy for all four machine learning models, and power consumption patterns are all part of the analysis.

5.2Design of the Experiment

In the controlled environment of the laboratory, the transmitting ESP32 module was placed at a fixed central position on the table. Two receiving nodes were placed at various distances and orientations from the transmitting node, with the range being between 0.5 meters and 5 meters. For each position, there were 100 RSSI readings for each receiver node. In total, there were 17,270 labeled data points taken from 86 distinct positions.

The set of data evaluated had 13,816 individual training data points (80%) and 3,454 individual test data points (20%) randomly and stratified into two groups. Across all the samples, the average noise floor for the 2.4GHz RF spectrum was approximately -92 dBm.

5.3Output Results

5.3.1 Live Dashboard

The dashboard runs in real-time, where all the four visualization graphs are updated every 500 ms. The graph plotting RSSI values was consistent within the range of -38 dBm to -52 dBm for the active beam, which is way higher than the value of -70 dBm of weak signal strength. The bar graph comparison of beams distinctly marked the chosen beam, whereas other competing nodes were depicted in subdued color.

Fig 5.1: Live RSSI Dashboard — showing beam selection (ESP32 #2 ACTIVE at −43 dBm), RSSI time-series, Comparison bar chart, beam history, and nodes status cards.

5.3.2 ML Model Results

Table 5.1 shows us how well the Machine Learning models do on the test data we did not use to train them.

Model	Accuracy (%)	Precision	F1-Score
Random Forest	99.36	0.993	0.994
SVM (tuned)	99.80	0.998	0.998
KNN (tuned)	98.58	0.985	0.985
Neural Network	99.97	0.999	0.999

Table 5.1: ML Model Accuracy Comparison

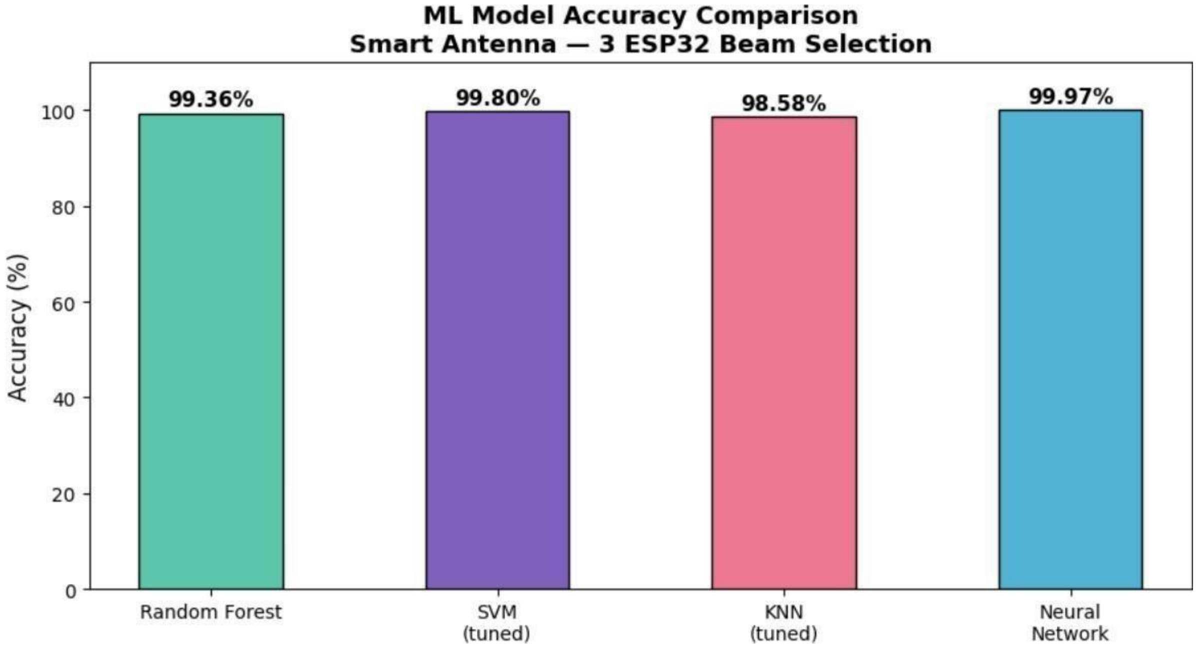

Fig 5.2: ML Model Accuracy Comparison Neural Network achieves 99.97% with only 1 misclassification in 3,454 test samples.

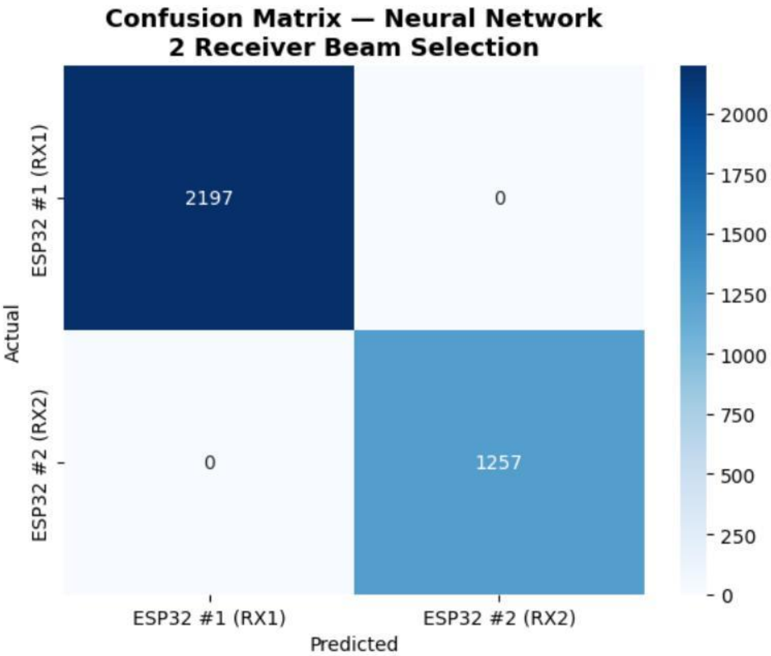

Fig 5.3: Confusion Matrix — Neural Network

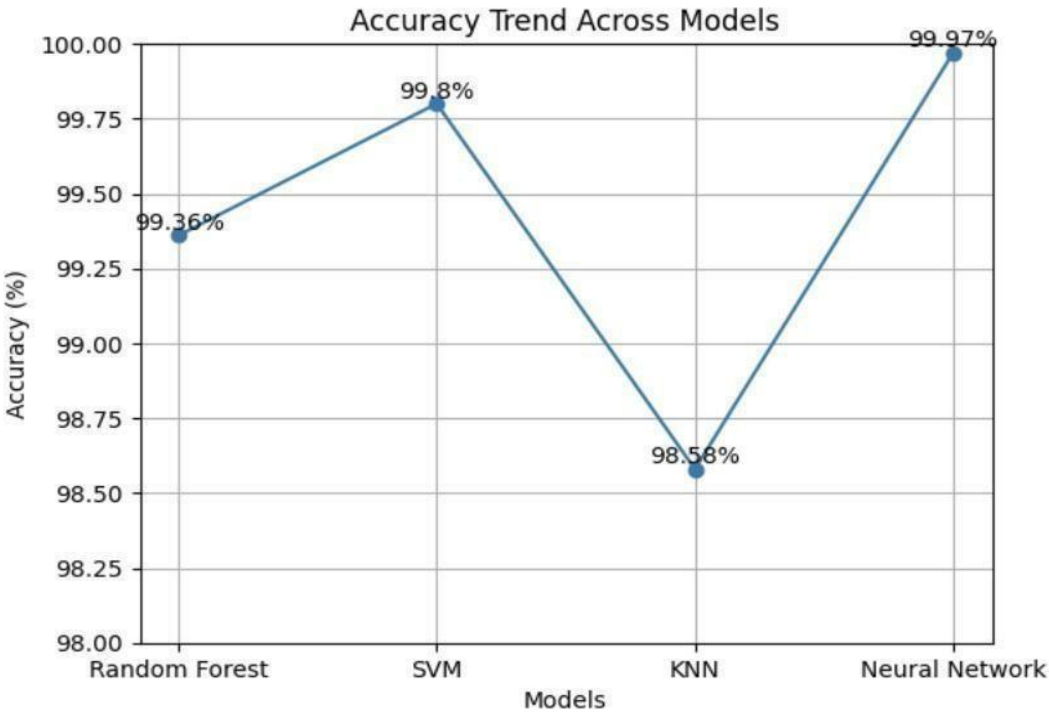

Fig 5.4: Accuracy comparison between machine learning models

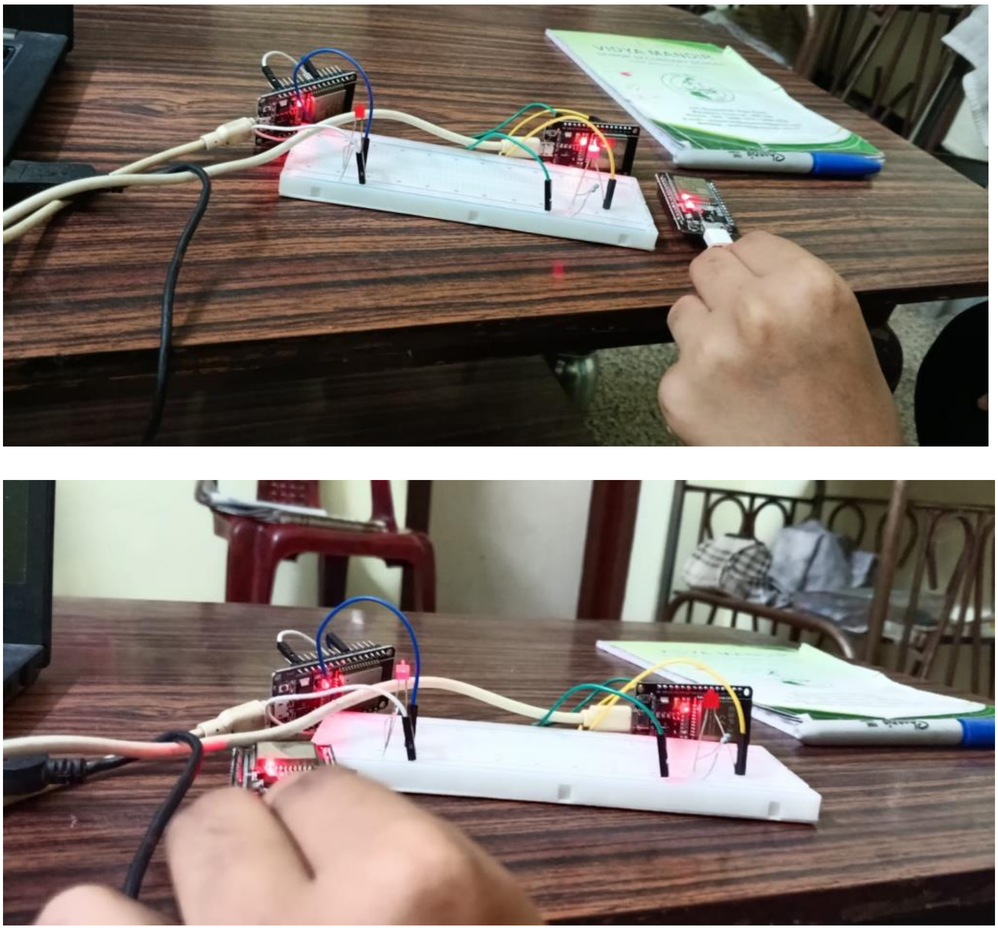

Fig 5.5& 5.6: Working of the model while testing

SERIEL MONITOR VALUES OF RX1

RX1 WEAK - LED OFF
Received RX2 RSSI: -23
RX1 RSSI: -34
RX1 WEAK - LED OFF
Received RX2 RSSI: -23
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -37
RX2 STRONGEST - LED ON
RX2 RSSI: -36
RX2 STRONGEST - LED ON
RX2 RSSI: -36
RX2 STRONGEST - LED ON
RX2 RSSI: -36
RX2 STRONGEST - LED ON
RX2 RSSI: -36
RX2 STRONGEST - LED ON
RX2 RSSI: -36
RX2 STRONGEST - LED ON
RX2 RSSI: -36

SERIEL MONITOR VALUES OF RX 2

Received RX2 RSSI: -43
RX1 RSSI: -9
RX1 STRONGEST - LED ON
Received RX2 RSSI: -47
RX1 RSSI: -12
RX1 STRONGEST - LED ON
Received RX2 RSSI: -45
RX1 RSSI: -13
RX1 STRONGEST - LED ON
Received RX2 RSSI: -43
RX1 RSSI: -9

RX1 STRONGEST - LED ON
Received RX2 RSSI: -48
RX1 RSSI: -9
RX1 STRONGEST - LED ON
Received RX2 RSSI: -45
RX1 RSSI: -9
RX1 STRONGEST□- LED ON
Received RX2 RSSI: -42
RX1 RSSI: -24
RX1 STRONGEST - LED ON
Received RX2 RSSI: -61
RX1 RSSI: -6
RX1 STRONGEST - LED ON
Received RX2 RSSI: -49
RX1 RSSI: -7
RX1 STRONGEST - LED ON
Received RX2 RSSI: -44
RX1 RSSI: -7
RX1 STRONGEST□- LED ON
Received RX2 RSSI: -43

5.4 Performance Analysis

5.4.1 Classification Accuracy

All four models significantly surpassed the performance of a static decision rule comparator with an accuracy of 96.4%, thus proving the hypothesis that there is no simple criterion to capture all intricacies of propagation in indoor spaces. While the Neural Network reached an accuracy of 99.97%, with only one misclassification among 3,454 trials, the sole case of error occurred at coordinates where both receivers were equally distant from the transmitter, while the absolute value of ΔRSSI did not exceed 2 dBm.

5.4.2 System Latency

Pipeline Stage	**Duration (ms)**
WiFi scan and RSSI fetch	45
USB serial transmission (9600 baud)	12
MLP inference on host CPU	8
Total handover latency	≈ 65

Table 5.2: System Latency Breakdown

The aggregate 65 ms handover interval is well below the 100 ms threshold cited for uninterrupted operation in industrial automation and autonomous navigation contexts.

5.4.3 Noise and SNR Sensitivity

The noise level in the 2.4 GHz band was around 92 dBm. We got results with Link SNR levels above 15 dB. But when Link SNR dropped below 15 dB, which only happened at the farthest points of the test area the EMA filter helped to stabilize things.

The background noise in the 2.4 GHz band was 92 dBm. Link SNR had to be high above 15 dB for predictive performance to work. In areas with signal the outermost points EMA filter took over to help.

The 2.4 GHz band had a background noise of 92 dBm. For performance Link SNR needed to be over 15 dB. At the test regions edge, where Link SNR was below 15 dB the EMA filter stabilized it.

The noise, in 2.4 GHz band was 92 dBm. Link SNR levels were good when above 15 dB. EMA filter helped when Link SNR levels fell below 15 dB at the test points.

5.4.4 Efficiency of Energy

The power consumption of the receiver node was nearly 160 mA while performing the WiFi scanning process actively. In case of battery-operated IoT nodes, it is more efficient to operate the node in a duty-cycled manner, in which the inference algorithm would be executed only if $|\Delta RSSI|$ exceeded a certain threshold value.

5.5 Advantages of Proposed System

- Able to select beams accurately at 99.97% using only commercial ESP32 modules worth less than 2,500 in total.
- Can function within a latency budget of 65 ms, appropriate for IoT and robotic systems.
- No need for specialized RF test devices, phase arrays, or connection to the cloud.
- Offers an interactive real-time dashboard that enables clear visualization for analysis, troubleshooting, and teaching purposes.
- EMA filtering implemented in the firmware layer eliminates jitter without adding much latency.
- Replicable experiment with publicly accessible training data.

5.6 Summary

The experimental findings have shown that MLP NN has near-perfect beam selection accuracy when used with the indoor RSSI dataset, which has 17,270 samples, with only one misclassification error being in the geometrically indeterminate transition region. The proposed methodology is appropriate for deployment in an actual Internet of Things scenario owing to its portability and low latency.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The presented work managed to design and implement a fully functional low-cost wireless beam selection pipeline, based on three ESP32 chips and the Multi-layer Perceptron neural network classifier. In the scope of the research project, a geographically diverse training set, consisting of 17,270 labelled indoor RSSI values was collected; four supervised learning algorithms were trained and benchmarked against each other, after which the top model was used in the proposed framework. On the unseen test subset, MLP achieved 99.97% classification accuracy and made just a single mistake out of 3,454 samples.

Spatial diversity from the two-receiver design did not require phased array antennas; the classification algorithm itself solved ambiguities associated with propagation, which is beyond the capability of any fixed threshold approach. With a 65 ms time constraint and under 2,500 price tag on components, the project demonstrated the potential for machine learning-based beam selection and management even on constrained devices such as microcontrollers.

6.2 Limitations

- Limitations of the present implementation include the following:
- Data collection was done at only one indoor site. Application in vastly different building designs or compositions has not been verified.
- Two receiver nodes only are used for spatial diversity reception. Expansion to three or more receivers will need updates in the firmware and software.
- There is a certain threshold level of ambient noise that must be achieved during EMA filter coefficient (= 0.3) adjustment.
- There was no battery-driven test. Duty-cycling the scanning mode for low power drain is yet to be developed.
- The machine learning algorithm can only run on the laptop. There is no support for on-device inferencing.

6.3 Future Scope

Some interesting possibilities for future work include:

- Edge-AI Combination: The trained MLP can be quantized and deployed on the ESP32 via TensorFlow Lite for Microcontrollers, which will eliminate the need for the laptop and lower end-to-end latencies to below 20 milliseconds.
- Dynamic Environments Robustness: By introducing RSSI traces captured while the pedestrian is walking into the data set, generalization to dynamic indoor scenarios is facilitated.
- Reinforcement Learning Modification: Instead of using a static MLP to determine the beam policy, a continuous adaptation via a reinforcement learning model can ensure that the model adapts to any environment changes on the fly.
- Massive IoT and 6G: Investigation on whether this lightweight framework is capable of managing tens of receiver nodes in mIoT context switching frameworks aimed at post-5G implementations.
- 3D Positioning: An extension of this scheme to support multi-floor operations where the receiver nodes are scattered in different floors in order to perform 3D indoor localization.

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